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Exploring Big Data on a Desktop
Putting Data into HDFS

Hadoop is a large scale open source storage and processing framework for data sets. This article builds upon the previous article in this series, 'Exploring Big Data on a Desktop: Getting Started with Hadoop' that appeared in last month's issue of OSFY. The focus, now, is on putting data into the system.

Once you have the basics of Hadoop functioning, the obvious question is: how does one put the data into the system? And the corollary to that is: how does one distribute the data between the various data nodes? Since the system can grow to thousands of data nodes, it is pretty obvious that you should not have to tell HDFS anything more than the bare minimum.

A simple test

Your first test can be to take a large text file and load it into HDFS. Start the Hadoop cluster as described in the previous article, with three nodes. On the first test, stop the datanode service on *h-mstr* and copy a large file on two slave nodes. (To create a text file for testing, take a hexdump of a 1GB tar —you should get a text file of around 3GB.)

```
$ ssh fedora@h-mstr
$ sudo systemctl stop hadoop-datanode
$ hdfs dfs -copyFromLocal  large_file.txt
```

You can verify that the data has been copied and how much resides on each node, as follows:

```
$ hadoop fs -ls -h large_file.txt
Found 1 items
-rw-r--r--   1 fedora supergroup      3.3 G 2014-08-15 12:25
large_file.txt
$ sudo du -sh /var/cache/hadoop-hdfs/
13M     /var/cache/hadoop-hdfs/
$ ssh -t fedora@h-slv1 du -sh /var/cache/hadoop-hdfs/
1.7G    /var/cache/hadoop-hdfs/
Connection to h-slv1 closed.
$ ssh -t fedora@h-slv2 du -sh /var/cache/hadoop-hdfs/
[fedora@h-mstr ~]$ ssh -t fedora@h-slv2 sudo du -sh /var/
cache/hadoop-hdfs/
1.7G    /var/cache/hadoop-hdfs/
Connection to h-slv2 closed.
```

The data file is distributed on *h-slv1* and *h-slv2*. Now, start the datanode on *h-mstr* as well. Run the above commands again, as follows:

```
$ sudo systemctl start hadoop-datanode
$ hdfs dfs -copyFromLocal  large_file.txt  second_large_file.txt
$ hadoop fs -ls -h
-rw-r--r--   1 fedora supergroup      3.3 G 2014-08-15 12:25
large_file.txt
-rw-r--r--   1 fedora supergroup      3.3 G 2014-08-15 12:33
second_large_file.txt
$ sudo du -sh /var/cache/hadoop-hdfs/
[fedora@h-mstr ~]$ sudo du -sh /var/cache/hadoop-hdfs/
3.4G    /var/cache/hadoop-hdfs/
$ ssh -t fedora@h-slv1 du -sh /var/cache/hadoop-hdfs/
1.7G    /var/cache/hadoop-hdfs/
Connection to h-slv1 closed.
[fedora@h-mstr ~]$ ssh -t fedora@h-slv2 sudo du -sh /var/cache/
hadoop-hdfs/
1.7G    /var/cache/hadoop-hdfs/
Connection to h-slv2 closed.
```

The new data seems to be stored on the *h-mstr* only. If you run the command from *h-slv2*, the data is likely to be stored on the *h-slv2* datanode only. That is, HDFS will try to optimise and store the data close to the origin. You can find out about the status of HDFS, including the location of data blocks, by browsing *http://h-mstr:50070/*.

The file you want to put in HDFS may be on the desktop. You can, of course, access that file using *nfs*. However, if you have installed the HDFS binaries on your desktop, you can copy the file by running the following command on the desktop:

```
$ HADOOP_USER_NAME=fedora hdfs dfs -fs hdfs://h-mstr/ -put
desktop_file.txt
```

The commands *-put* and *-copyFromLocal* are synonyms.

Replication

You should stop the Hadoop services on all the nodes. In */etc/hadoop/hdfs-site.xml*, on each of the nodes, change the value of *dfs.replication* from 1 to 2 and the data blocks will be replicated on one additional datanode.

```
<property>
```